

Scan Your Local Network for Open Ports

Objectives:

Learn how to:

- Discover devices in your network.
- Find open ports.
- Understand network exposure and security risks.

Tool Required:

1. Nmap(mandatory).
2. Wireshark(optional).

You can install them for:

- nmap.org
- wireshark.org

Procedure:

Step1: Download and install Nmap from the official website.

During installation:

- Keep default setting.
- Install Npcap if asked.

Step2: Find Your Local IP Range.

On Windows

Open command prompt and type: ipconfig

```
Microsoft Windows [Version 10.0.26200.8457]  
(c) Microsoft Corporation. All rights reserved.
```

```
C:\Users\manjeerika>ipconfig
```

```
Windows IP Configuration
```

```
Wireless LAN adapter Local Area Connection* 1:
```

```
Media State . . . . . : Media disconnected  
Connection-specific DNS Suffix . :
```

```
Wireless LAN adapter Local Area Connection* 2:
```

```
Media State . . . . . : Media disconnected  
Connection-specific DNS Suffix . :
```

```
Wireless LAN adapter Wi-Fi:
```

```
Connection-specific DNS Suffix . :  
IPv6 Address. . . . . : 2405:e3c0:6:f7:fd4f:1780:ad3e:9d3c  
Temporary IPv6 Address. . . . . : 2405:e3c0:6:f7:ad99:37ca:da01:2ff6  
Link-local IPv6 Address . . . . . : fe80::d8d5:f84a:1a90:5ddb%6  
IPv4 Address. . . . . : 192.168.1.7  
Subnet Mask . . . . . : 255.255.255.0  
Default Gateway . . . . . : fe80::56a2:45ff:fe0b:f9b8%6  
192.168.1.1
```

Step3: Scan the Network Using Nmap

Open command prompt and run: nmap -s

```
(manjeerika@kali) ~$  
~$ nmap -s  
Nmap 7.95 ( https://nmap.org )  
Usage: nmap [Scan Type(s)] [Options] {target specification}  
TARGET SPECIFICATION:  
  Can pass hostnames, IP addresses, networks, etc.  
  Ex: scanme.nmap.org, microsoft.com/24, 192.168.0.1; 10.0.0-255.1-254  
  -iL <inputfilename>: Input from list of hosts/networks  
  -iR <num hosts>: Choose random targets  
  --exclude <host1[,host2][,host3],...>: Exclude hosts/networks  
  --exclude-file <exclude_file>: Exclude list from file  
HOST DISCOVERY:  
  -sL: List Scan - simply list targets to scan  
  -sn: Ping Scan - disable port scan  
  -Pn: Treat all hosts as online -- skip host discovery  
  -PS/PA/PY[portlist]: TCP SYN, TCP ACK, UDP or SCTP discovery to given ports  
  -PE/PP/PM: ICMP echo, timestamp, and netmask request discovery probes  
  -PO[protocol list]: IP Protocol Ping  
  -n/-R: Never do DNS resolution/Always resolve [default: sometimes]  
  --dns-servers <serv1[,serv2],...>: Specify custom DNS servers  
  --system-dns: Use OS's DNS resolver  
  --traceroute: Trace hop path to each host  
SCAN TECHNIQUES:  
  -sS/sT/sA/sW/sM: TCP SYN/Connect()/ACK/Window/Maimon scans  
  -sU: UDP Scan  
  -sM/sF/sX: TCP Null, FIN, and Xmas scans  
  --scanflags <flags>: Customize TCP scan flags  
  -sI <zombie host[:probeport]>: Idle scan  
  -sV/sZ: SCTP INIT/COOKIE-ECHO scans  
  -sO: IP protocol scan  
  -b <FTP relay host>: FTP bounce scan  
PORT SPECIFICATION AND SCAN ORDER:  
  -p <port ranges>: Only scan specified ports  
    Ex: -p22; -p1-65535; -p U:53,I:137,T:21-25,80,139,8080,S:9  
  --exclude-ports <port ranges>: Exclude the specified ports from scanning  
  -F: Fast mode - Scan fewer ports than the default scan  
  -r: Scan ports sequentially - don't randomize  
  --top-ports <number>: Scan <number> most common ports  
  --port-ratio <ratio>: Scan ports more common than <ratio>  
SERVICE/VERSION DETECTION:  
  -sV: Probe open ports to determine service/version info  
  --version-intensity <level>: Set from 0 (light) to 9 (try all probes)  
  --version-light: Limit to most likely probes (intensity 2)  
  --version-all: Try every single probe (intensity 9)  
  --version-trace: Show detailed version scan activity (for debugging)
```

Step4: Observe the Results

You will see:

- IP address.
- Hostnames.
- Open ports.
- Running services.

Step5: Note Down the Information.

IP address	Open ports	Services
192.169.1.1	80	HTTP
192.168.1.1	443	HTTPS
192.168.1.5	22	SSH

Step6: Research the Ports.

Common ports:

Port	Service
20/21	FTP
22	SSH
23	Telnet
25	SMTP
53	DNS
80	HTTP
110	POP3
139	NetBIOS
443	HTTPS

Step7: Identify Security Risks.

Examples:

- Open Telnet port -insecure communication.
- Unused open ports -attack possibility.
- Weak services -vulnerability risk

Step8: Save the Scan Result.

Optional Wireshark Activity

1. Open Wireshark.
2. Select Wi-Fi / Ethernet interface.
3. Start Capture.
4. Run Nmap scan.
5. Stop capture.
6. Observe packets.

Tool Required:

- Nmap.
- Wireshark.

Procedure:

Step1: Install Wireshark.

Step2: Find your local IP range.

Step3: Scan the network using Wireshark.

Step4: Observe the Results.

Step5: Note down the information.

Step6: Research the ports.

Step7: Identify security risks

Step8: Save the scan result.

Outcomes:

Learned the basics of port scanning using Nmap packet capturing using Wireshark network reconnaissance understanding open ports and services basic network security concepts.

Conclusion:

Successfully scanned the local network using Nmap and identified active devices and open ports.

Important Note:

Do this only on:

- Your own system.
- Your home/local network.
- Authorized environments

Do not scan public or unknown networks without permission.